

Kenneth Zhang

RESEARCHER | LEARNING-AUGMENTED
OPTIMIZATION

Guangzhou / Macau / Zhuhai
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PROFILE

Undergraduate researcher at the Faculty of Data Science, City University of Macau. I study how learned structural priors can guide classical optimization solvers for constrained routing and intelligent transportation.

EDUCATION

City University of Macau | Faculty of Data Science

Bachelor of Engineering in Computer Science and Technology | Sep 2023 – Jun 2027

GPA: 3.4/4.0 | Expected Graduation: 2027

Selected Coursework: Data Structures and Algorithms, Optimization Theory, Probability and Statistics, Reinforcement Learning, and Deep Learning

Language

Mandarin: Native

Cantonese: Native

English: Professional working proficiency; able to read academic papers and technical documentation

Research Interests

Neural Combinatorial Optimization (NCO) / Constrained Reinforcement Learning / Route Planning / Large Language Model Reasoning

- Open to research collaborations and algorithm-related opportunities in Macau, Zhuhai, Guangzhou, and remotely.

RESEARCH INTERNSHIP EXPERIENCE

Research Assistant | iLSCA Lab, HKUST(GZ)

Jun 2026 – Aug 2026 | Guangzhou, China

Manuscript under review at IEEE T-ITS | Co-author

Group-Constrained TSP / Cluster-Aware Attention

Under Review | Submitted: 21-May-2026

- Co-authored A Unified Group-Aware Deep Reinforcement Learning Framework for Group-Constrained TSPs. Contributed to the design of Cluster-Aware Attention, comparative experiments, and the reproduction and tuning of LKH3 and ALNS baselines.

First-Author Manuscript in Preparation

Cluster-Aware Learned Edge Priors for Guiding LKH Solvers

Target: IEEE TEVC Special Issue | EC Meets ML for CO

- Developed a learning-augmented optimization framework that integrates cluster-aware edge priors with Population-LKH. The method uses Transformer-based edge prediction, prior sparsification, and a guarded selector to guide LKH across BWTSP and CTSP-d, with comparisons against Population-LKH, ALNS, Gurobi, and a NeuroLKH-style control.